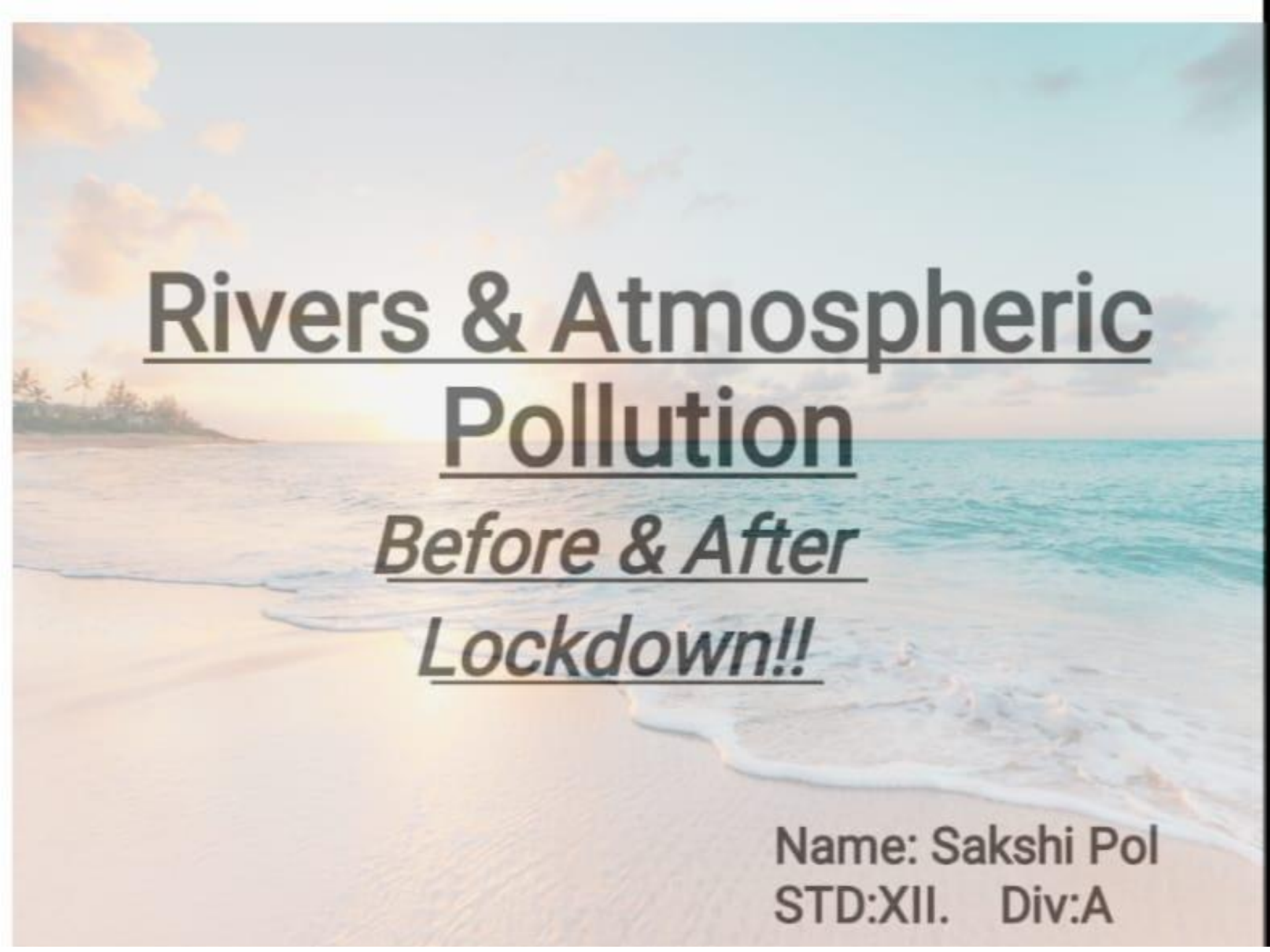
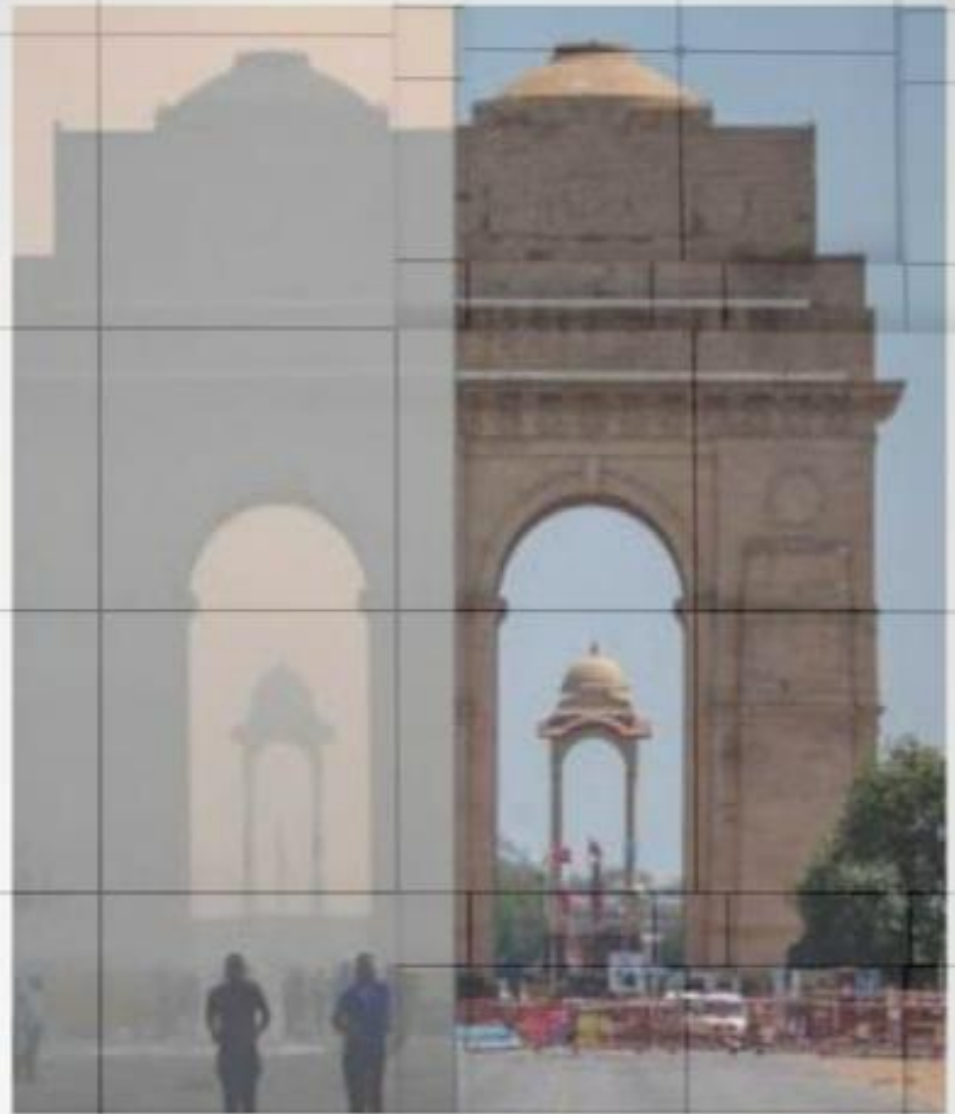




Rivers & Atmospheric Pollution *Before & After* *Lockdown!!*

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STD:XII. Div:A

***Atmospheric
Pollution
Before & After
Lockdown***



***River Pollution
Before & After
Lockdown***



CLIMATE CHANGE

Climate change

So today, I am going to share something that you must have been witnessing around you during this lockdown.

As you all can see how the sky has turned blue, the air has become cleaner. When we look up at the night sky, we actually see the stars.

Atmospheric Pollution

- As the coronavirus pandemic swept through Asia, many countries imposed strict lockdowns to stop the disease from spreading, especially in major cities.
- After lockdown the air has become so clear that the Himalayan peaks are visible from Jalandhar in Punjab, situated 140kms away.
- The residents of Jalandhar say that for the first time in 20-30 years, the air has become so clean that they are able to see the Himalayan peaks from Jalandhar.
- Such news reports are pouring in from worldwide. The air has become cleaner, the water has become clearer, the pollution levels have fallen.

NITROGEN DIOXIDE

Nitrogen Dioxide(NO_2)

•Before :-

- Nitrogen dioxide is a toxic gas that is emitted from the engines of cars, buses, trucks and from factories.
- The WHO states that if this gas is more in concentration than 200 micrograms per cubic meter, then it can cause an enormous inflammation in your breathing pipe which lead to problems like asthma.

•After:-

- Now, due to this lockdown, the number of cars playing on the roads have become less factories have shut down.
- Hence, in the cities all over the world the concentration of nitrogen dioxide in the air has fallen drastically.

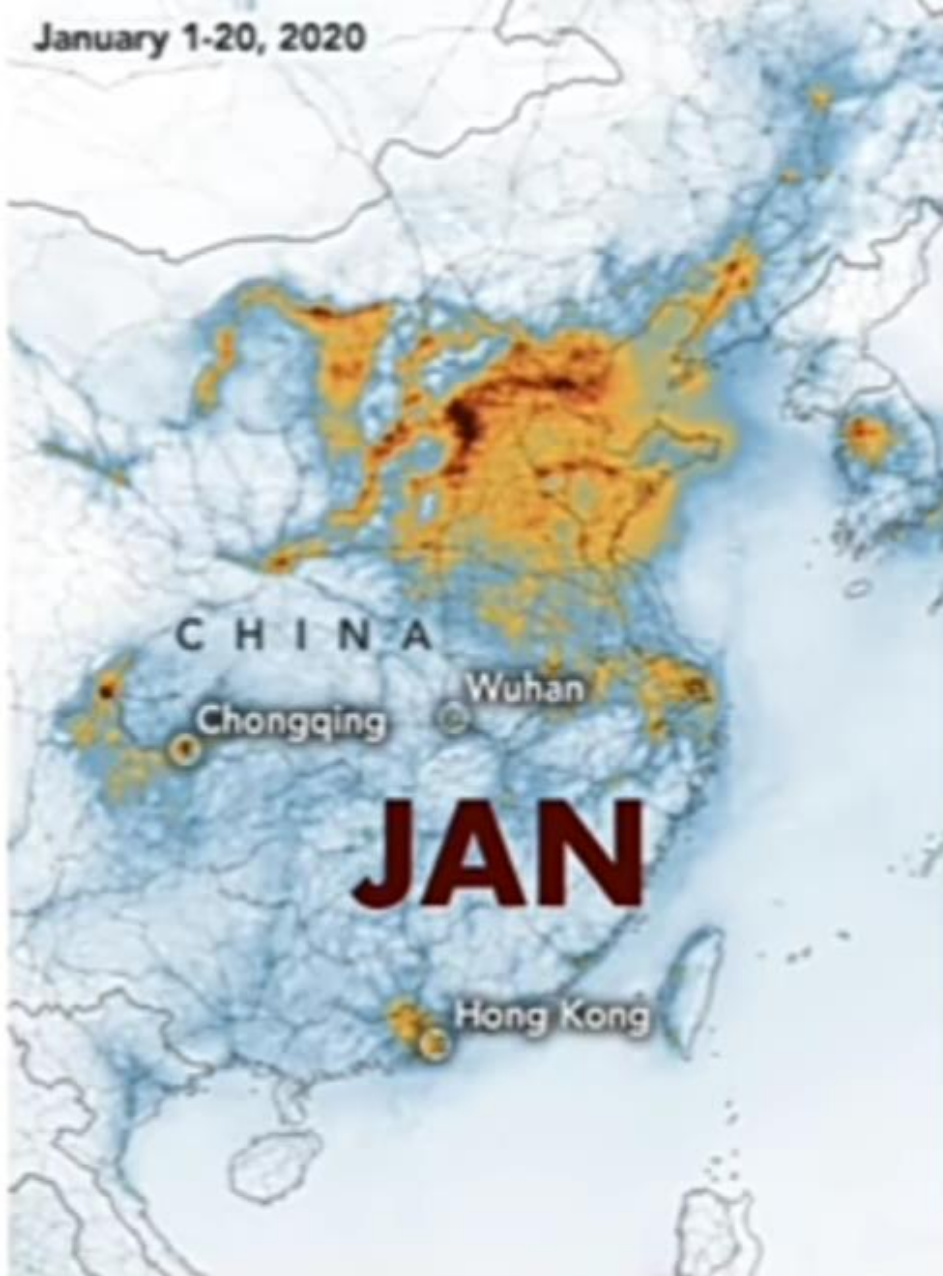
Examples:-

First example in China: Look at the levels in the month of January and February and the look at how much it has fallen in February and till now.

Exactly same thing was witness in the cities of Europe in Spain in UK, in Italy.



January 1-20, 2020



February 10-25, 2020



Mean Tropospheric NO₂ Density ($\mu\text{mol}/\text{m}^2$)



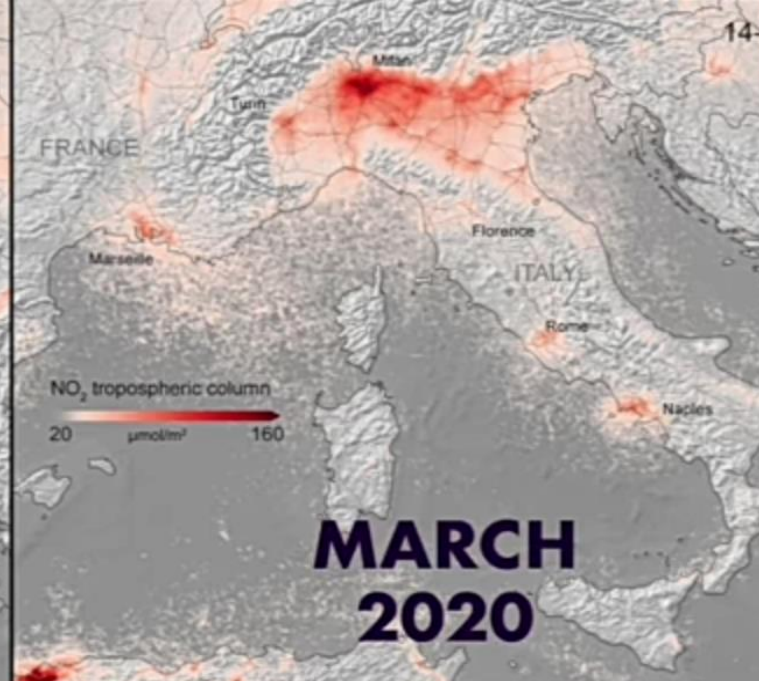
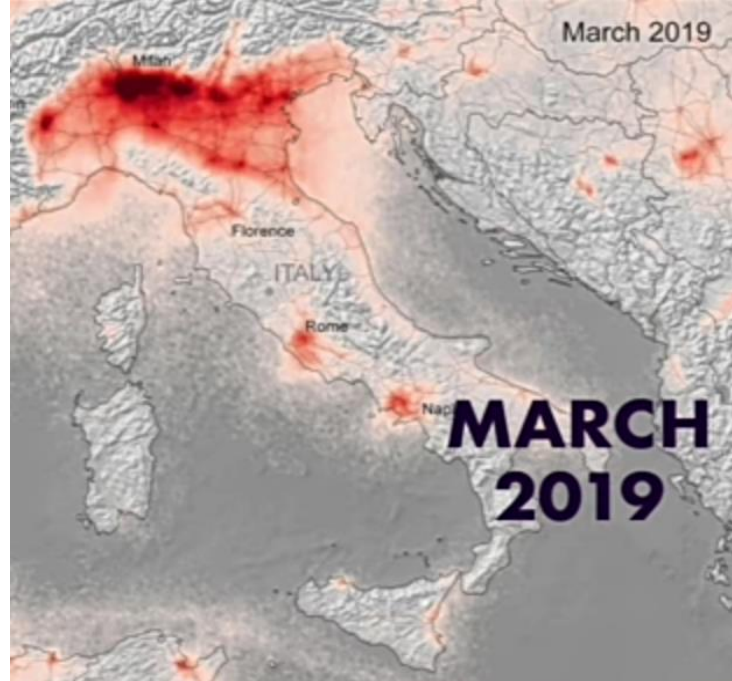
Example:-

Take example of of Italy:
Notice how much of
NO₂ was present in
Italy in March 2019 and
how much it got reduce
in March 2020.



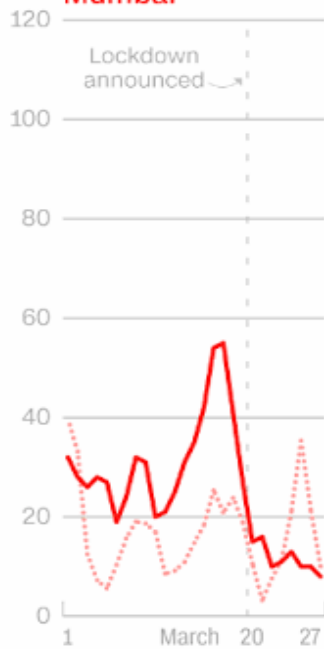
Exactly the same thing
was witnessed in Indian
cities too look at Delhi,
Mumbai, Bangalore,
Chennai, Hyderabad.

*So you can observe how
the NO₂, concentration
have fallen in comparison
to 2019 in these cities and
comparison in last few
months.*

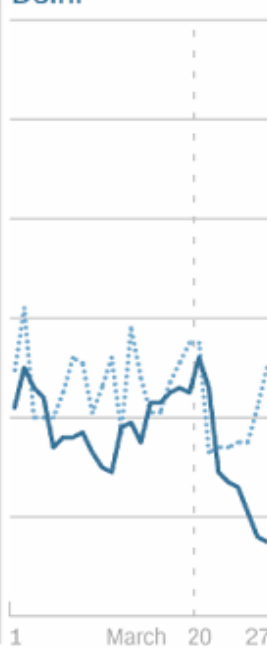


— 2020 2019

Mumbai



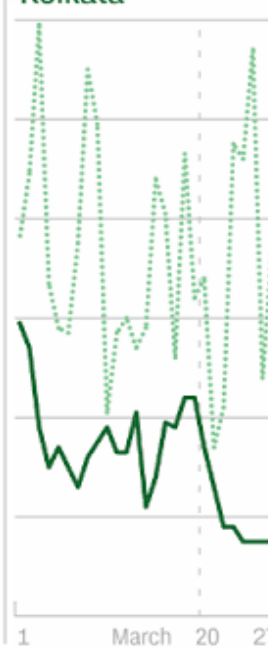
Delhi



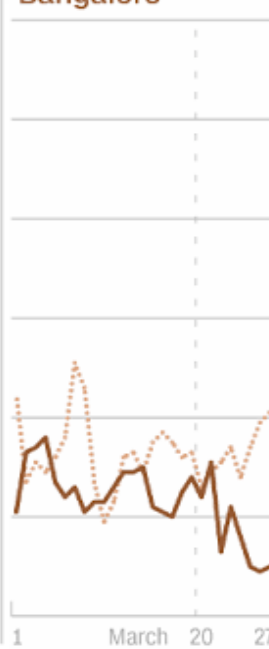
Chennai



Kolkata



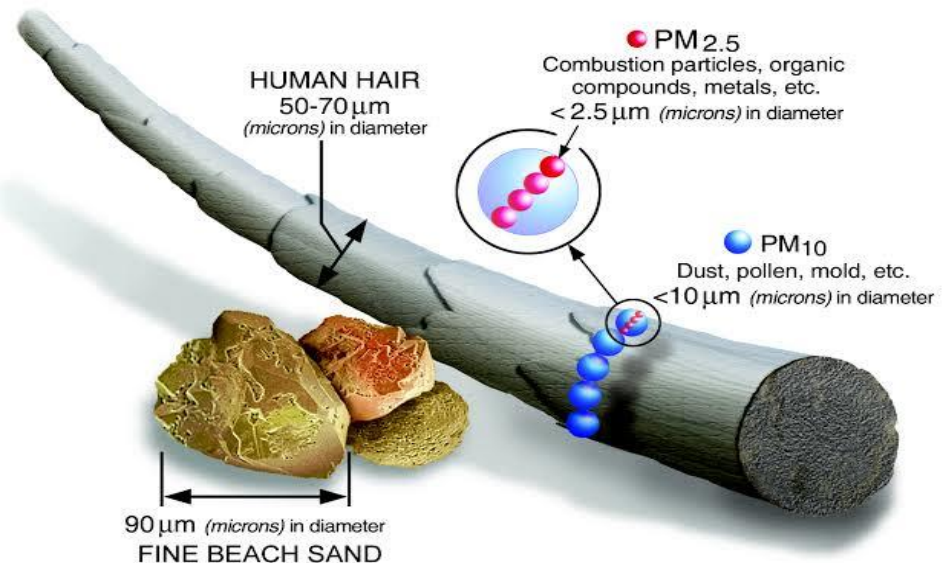
Bangalore



PM 2.5

PM 2.5 (Before)

- Particulate matter 2.5 or PM 2.5. It is one of the most harmful forms of air pollution.
- It is included in the category of group 1 carcinogenic and it is so small that it can travel from your lungs into your blood stream which will not only cause respiratory problems but also heart attack which can also cause premature deaths.
- It is so deadly that WHO has estimated that every year worldwide that more than 4 million people are killed due to PM 2.5 caused heart diseases, strokes, lung cancer, chronic lung diseases and respiratory infection.



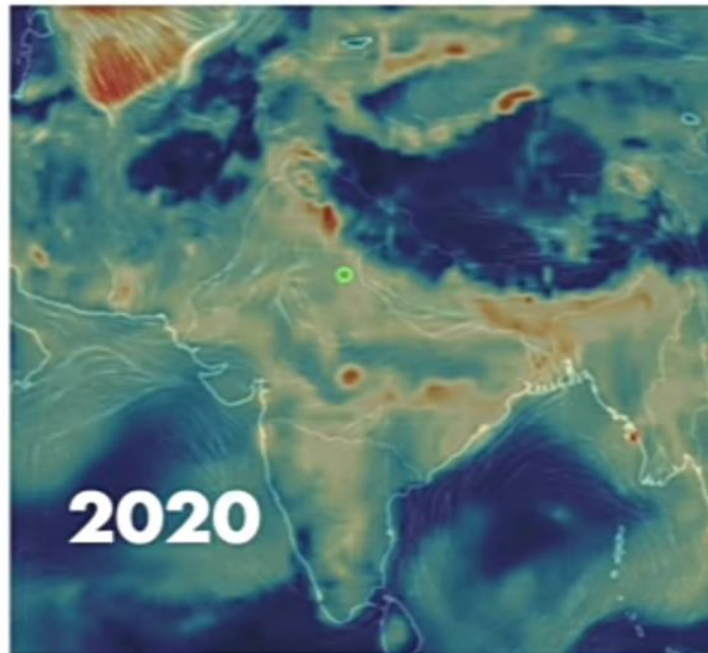
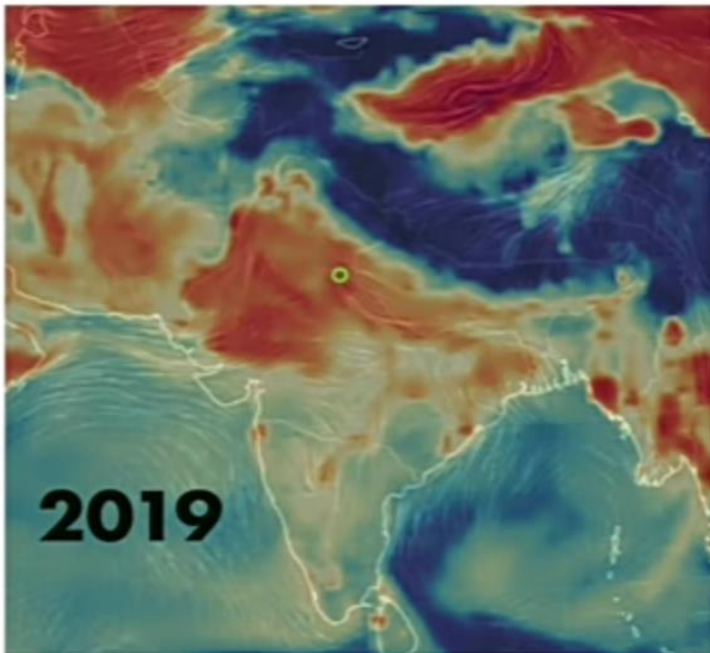
PM 2.5 (After)

- Now, due to lockdown, even the PM 2.5 levels have reduced worldwide.
- Have a look at India's map.



Instantaneous PM_{2.5} levels over India on 7th April 2019 and 2020

Data Source- earth.nullsch



PM 2.5 (After)

- We saw the concentration of the PM 2.5 level in the entire country on 7th April 2019 and what was the concentration (of the PM 2.5 levels) on 7th April, 2020.
- It's has depleted by a significant amount.
- This is the reason why, these days, you are able to view the clear blue sky and stars at night in different cities across the country.
- Take a look at this photos of Mumbai:



*Examples:-

- You would not even be able to recognize. It seems as if it is a city from a foreign country.
- This tells us how much difference does air pollution make in the beauty of a city.



•But most of the people in our country are not concerned with such issues.

•Now if a mindir-masjid issue is raised,than hundreds of people would rise up to riot.

•But if air pollution is talked about, nobodys voice is heard.



Is clean air saving lives?

- A question might arise in your minds here that if PM 2.5 levels have reduced so drastically under lockdown, then the number of people dying of air pollution would also have become less.
- A scientist from Stanford University has done a research regarding this he collected the data of the PM 2.5 levels in Chinese cities and compared it to the mortality rates and observed what impacts would falling PM 2.5 levels would have on the mortality rates.

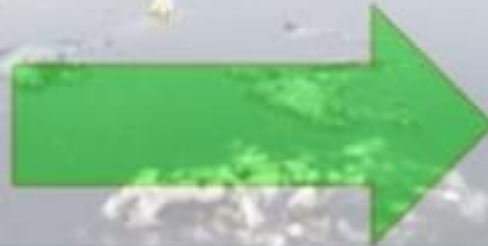
Conclusion for the question...

- And the research concluded that the falling levels of pollution in China in just this one month is more than 77,000 lives which have been saved due to the decrease in air pollution.
- Its such a massive number. The total number of deaths due to the COVID-19 worldwide is around 75,000.
- More than that number of lives have been saved just within China due to decrease in air pollution.

WATER POLLUTION

Water Pollution

- Not only the air, but the water of our Rivers is also becoming clearer and cleaner.
- Raghav Chadda, the Delhi Jal Board Vice Chairman says that due to the stoppage of the industrial waste has definitely caused a positive impact on the water quality of Yamuna River.
- The best example of this can be seen in photos.



Rivers Before and After

● This is the Yamuna River in Delhi. These photos have been captured from Delhi and Noida.

● It is quite unbelievable to see because compare it to how the Yamuna River appears normally.

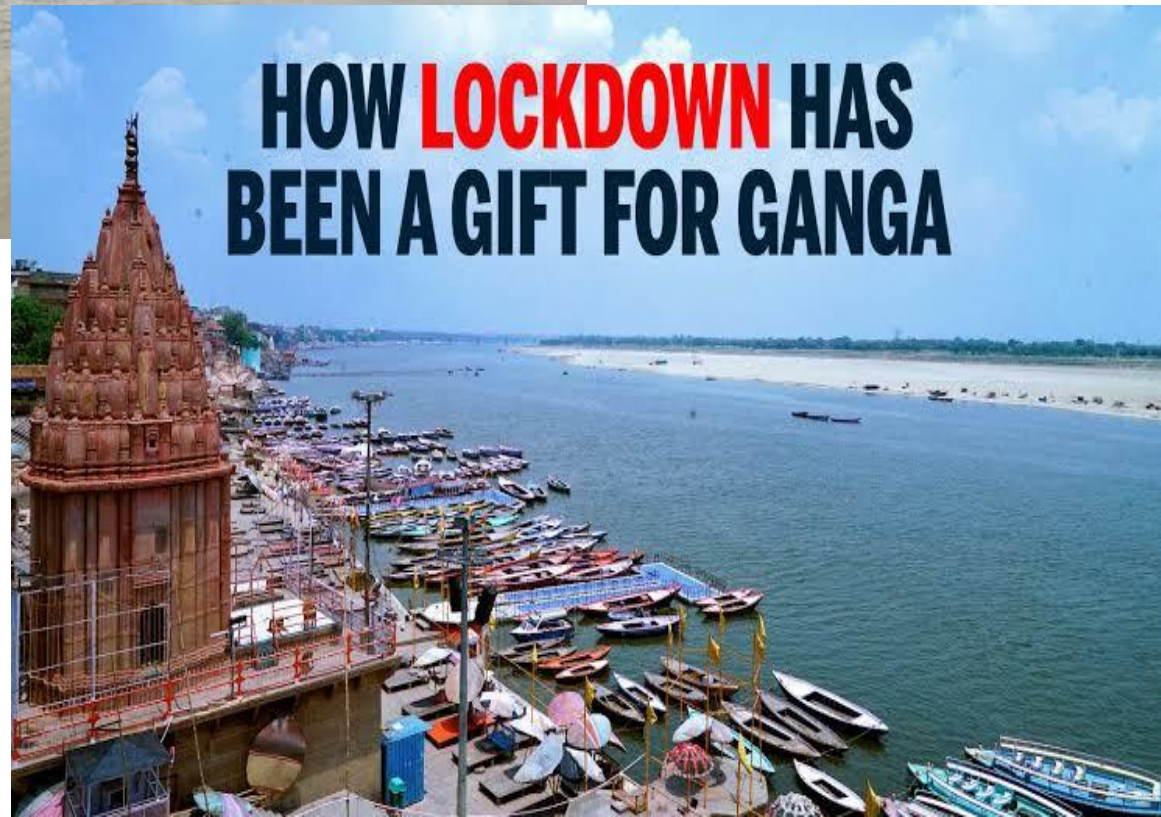




Ganga river

- This is the condition not only for the Yamuna but also of the Ganga river.
- Dr. P.K Mishra, an IIT Professor in BHU states that a 40-50% improvement has been seen in the water quality of the Ganga.
- Maybe we should ensure such lockdowns every year for a week so that the entire population of the country realizes and sees how clear can the air and water become they can catch a glimpse of the potential of where our country can reach.





#Opinion:-



The photos show nothing but the future potential. If the government constructs sewage treatment plants in the right manner and make strong regulation for the companies and industries to treat their waste themselves, then this can happen and our Rivers can appear this way.

THANK
YOU